

# Qwen3.8-27B Agentic Search Eval on Private Data

A head-to-head replacement decision against Qwen3.6-35B-A3B, on one NVIDIA L4, over a corpus that never leaves the premises.

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THIS DECK



# The replacement decision

Qwen3.8 was released last week and has been widely promoted since. For agentic search on my own hardware I serve **Qwen3.6-35B-A3B** by default, and I have spent months tuning how it is served. The public reception of a new model does not tell me whether to replace it.

In agentic search over my own documents, how much better is **Qwen3.8-27B** than **Qwen3.6-35B-A3B**?

the question this evaluation answers

I evaluate two specific backends that I have already deployed and measured on the same card. A public leaderboard cannot answer the question, because the documents I search are private and the evaluation has to be built from them.

# The two serving backends

## Qwen3.6-35B-A3B-MTP



hanxiao

Revert README agent prompt block

d501a47 · 5 days ago

37 Commits

|                    |                                                               |              |
|--------------------|---------------------------------------------------------------|--------------|
| .github/workflows  | Publish prebuilt decode-optimized image to GHCR via Git...    | 2 months ago |
| docs               | optimization-frontier: measured IQ4_XS speed AND qual...      | 2 months ago |
| patches            | Fix audit findings: malformed patch headers + README ra...    | 2 months ago |
| scripts            | Optional pre-baked tooling image: cold start ~3.5min -> ~...  | 2 months ago |
| Dockerfile         | Reconcile tok/s numbers to one signal; refresh headline fr... | 2 months ago |
| README.md          | Revert README agent prompt block                              | 5 days ago   |
| docker-compose.yml | Default --ctx-size 56320 everywhere (ECC off is the assu...   | 2 months ago |

README

### Qwen3.6-35B-A3B-MTP on NVIDIA L4 24GB

(Q4\_K\_XL) + MTP speculative decoding on a single NVIDIA L4 24GB — GCP g2-standard-8 — via the official llama.cpp Docker image. Decode-optimized to ~91-99 tok/s (min ~91 chat, max ~99 math), lossless (full GPU residency + ECC-off).

gcp

gguf

gpu-inference

inference-optimization

large-language-models

llama-cpp

llm

local-llm

mixture-of-experts

moe

multi-token-prediction

nvidia-l4

quantization

qwen

qwen3

speculative-decoding

Readme

sparse MoE · 35B total, ~3B active · 8 of 256 experts per token · UD-Q4\_K\_XL ~21.3 GiB resident · MTP head baked in

91-99

tok/s decode, chat minimum to math maximum

56,320

token context

## Qwen3.8-27B-UD-Q4\_K\_XL



hanxiao

Serve on the VPC only, and add a restart path

d4f3984 · 16 hours ago

8 Commits

|            |                                                               |              |
|------------|---------------------------------------------------------------|--------------|
| scripts    | Serve on the VPC only, and add a restart path                 | 16 hours ago |
| .gitignore | Qwen3.8-27B UD-Q4_K_XL on a single NVIDIA L4                  | 2 days ago   |
| LICENSE    | Fix repo metadata: description, topics, licence, stale com... | 2 days ago   |
| README.md  | Serve on the VPC only, and add a restart path                 | 16 hours ago |

README Apache-2.0 license

### Qwen3.8-27B · UD-Q4\_K\_XL · NVIDIA L4

Qwen3.8-27B serves at 23 tok/s decode over a 104,192-token context on a single NVIDIA L4 24 GB, using the [Unsloth Dynamic v3.0 UD-Q4\\_K\\_XL GGUF](#) and llama.cpp with MTP speculative decoding. That context is the largest the card will serve at this quantization.

Q4\_K\_XL GGUF at 23 tok/s over a 104,192-token context on a single NVIDIA L4 24GB, llama.cpp + MTP speculative decoding. One-command GCP provisioning, tuning sweep, quality verification, and a binary-searched context ceiling.

benchmark

gcp

gguf

llama-cpp

llm-inference

local-llm

mtp

nvidia-l4

quantization

qwen

qwen3

speculative-decoding

Readme

Apache-2.0 license

Activity

0 stars

hybrid attention · 64 layers, full\_attention\_interval 4, so 16 hold a KV cache · native MTP head · positions to 262,144

23

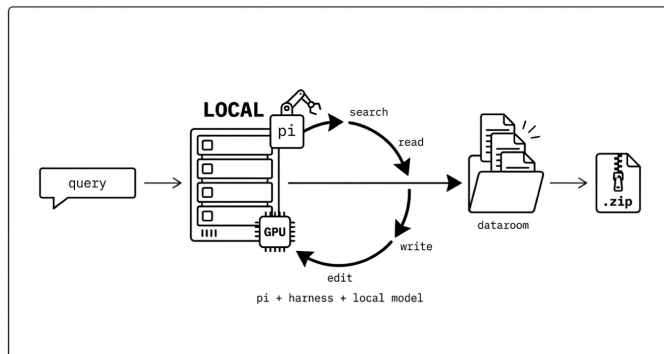
tok/s decode, binary-searched ceiling

104,192

token context, the largest the card serves at this quantization

# dataroom and searchbox

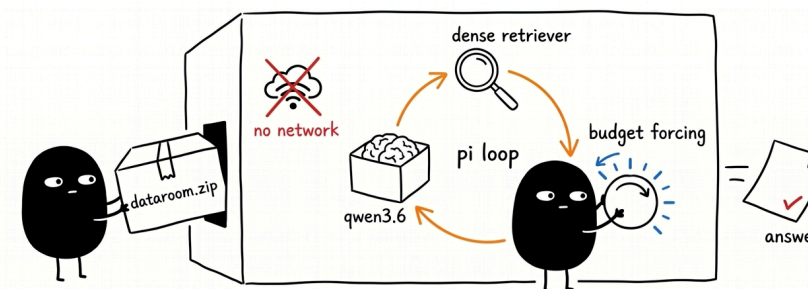
## dataroom



A local model runs **search, read and write** in a loop until the knowledge is dumped into one fully cited .zip on disk. It stops at a measured coverage floor.

[dataroom.jina.ai](https://dataroom.jina.ai)

## searchbox



An **airgapped** testbed for search as test-time compute. One local model is locked in with a single .zip dataroom and no web access, so every answer is composed from local tools over what is in the box.

[hanxiao.io/searchbox](https://hanxiao.io/searchbox)

# The private corpus

| GROUP              | FILES | DETAIL                                              |
|--------------------|-------|-----------------------------------------------------|
| <b>papers</b>      | 20    | six years of published work, 2023-07 to 2026-07     |
| <b>model-cards</b> | 29    | one spec sheet per released model                   |
| <b>blog</b>        | 161   | tech 104 / press 28 / insights 14 / events 8 / kb 7 |
| <b>api</b>         | 3     | OpenAPI spec, usage guide, model selection guide    |
| <b>faq</b>         | 2     | 163 question-answer pairs over 14 sources           |
| <b>site</b>        | 2     | 27 page namespaces, 2,297 English strings           |
| <b>legal</b>       | 1     | terms, privacy, security, DPA                       |

Snapshot 2026-08-13, 218 files.

7,777

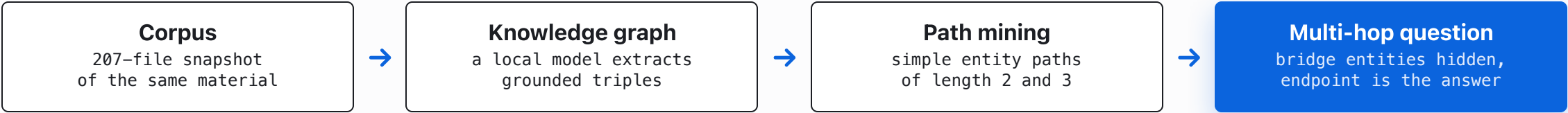
chunks indexed over 210 documents, median 91 units, longest 195 units at 2.1 KB

768

dimensions, jina-embeddings-v5-text-nano, L2-normalized

Six years of my own papers sit alongside the website material. Each released model carries both a paper and a release post, framed differently, and that overlap is what makes multi-hop questions possible over this corpus.

# Verifier construction



A path of length  $k$  is a chain of  $k$  corpus-grounded facts, so path length sets the difficulty. The question names the starting entity and hides the bridges, which forces a solver to discover them. A question enters the set when a closed-book probe fails on it and at least one retrieval-equipped solver reaches the answer, so the set measures search over the corpus. The graphs were built over an earlier 207-file snapshot of the same material, and the 218-file snapshot is what the agentic harness indexes today.

|                                                                    |                                                           |                                          |                                                      |
|--------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------|------------------------------------------------------|
| <b>10</b><br>graphs built over that corpus by the same local model | <b>18,615</b><br>edges audited, every edge of every graph | <b>40</b><br>questions sampled per graph | <b>2 and 3</b><br>hop counts the question set covers |
|--------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------|------------------------------------------------------|

# Evaluation design

|                 |             | CLOSED-BOOK                                       | SINGLE-SHOT BM25         | AGENTIC BM25, 4 ROUNDS   |
|-----------------|-------------|---------------------------------------------------|--------------------------|--------------------------|
| Qwen3.8-27B     | 104,192 ctx | control that removes items answerable from memory | top-5 passages, one shot | the condition under test |
| Qwen3.6-35B-A3B | 56,320 ctx  | control, same corpus                              | top-5 passages, one shot | the incumbent            |

## HELD FIXED

The corpus, the question set, the retrieval index, the grading procedure and the four-round cap. The agent runs in the airgapped loop with one retrieval tool over embeddings.npz and chunks.jsonl, plus stock file access. Only the served model changes across rows.

## SCORING

The answer held at the end of every turn is captured and graded by the same local judge, so accuracy is reported against turn index and against cumulative fresh-prefill input tokens. Turns are what a person reads and fresh-prefill tokens are what the GPU pays.

**A private corpus, a private verifier,  
and **one low-budget GPU**  
decide which model I serve.**

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THE HARNESS

